

List Directory Contents (1)

* Try to run 'ls -l'

```
(jiangjunmin@eex001 ~)$ ls -l
total 2292
-rw-r--r-- 1 jiangjunmin jiangjunmin 1262 Jul 6 13:59 bashrc usr
-rw-r--r-- 1 jiangjunmin jiangjunmin 75739 Aug 2 13:24 CDS.log
-rw-r--r-- 1 jiangjunmin jiangjunmin 2246498 Jul 16 13:04 CDS.log.1
drwxr-xr-x 7 jiangjunmin jiangjunmin 74 May 11 19:02 cserver
drwxr-xr-x 2 jiangjunmin jiangjunmin 6 Apr 30 16:31 Desktop
drwxr-xr-x 10 jiangjunmin jiangjunmin 210 Sep 8 19:19 Downloads
drwxr-xr-x 27 root root 4096 May 11 14:48 jiangjun
-rw-r--r-- 1 jiangjunmin jiangjunmin 467 Jul 9 01:11 Makefile
drwxr-xr-x 6 jiangjunmin jiangjunmin 92 Jul 21 21:47 project
drwxr-xr-x 21 jiangjunmin jiangjunmin 4096 Aug 1 22:41 simulation
-rwxr-xr-x 1 jiangjunmin jiangjunmin 40 May 11 19:17
```

* Type field (first character)

* **d** the entry is a directory;

- * **l** the entry is a symbolic link;
- * **b** the entry is a block special file;
- * **c** the entry is a character special file;
- * **-** the entry is an ordinary file;

* **cd [directory]**

* **cd Full Path Names**

* **cd Relative Path Names**

* **cd = cd ~ = cd \$HOME**

* **cd ~thomson**

Permission in Numeric

* **chmod [options] file**

using numeric representations for permissions:

```
r=4
w=2
x=1
Total: 7
```

* **chmod 7 7 7 filename**

user group others

* Gives user, group and others the **r, w, x** permissions by this representation

* **chmod 750 filename**

- * gives the user read, write, execute
- * gives group members read, execute
- * gives others no permissions

* **chmod 640 filename**

- * gives the user read, write
- * gives group members read
- * gives others no permissions

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File Manipulation Commands

-f 强制覆盖 **-i** 覆盖前询问
-p 保留源文件属性 **-R** 递归复制目录

* **cp** copy file

* **cp [-flRp] source_file target_file**

* **mv** move (or rename) file

* **mv [-fi] source target_file**

* **mv [-fi] source ... target_dir**

* **rm** remove (delete) a file (directory)

* **rm [-f] [-i] file ...**

* **rm -R [-f] [-i] dirname ... [file ...]**

* Meaning of each option here

grep/egrep/fgrep - search the argument for all occurrences of the search string (get REP)

* The **grep** utility is used to search for regular expressions in Unix files.

* **fgrep** searches for exact strings (not pattern).

* **egrep** uses "extended" regular expressions.

grep regexp file

* **grep [static/*.*]** static 字符, /*.* 找.c文件

* **grep [/*.*]** 行首是井号的文件

* **man ls** **grep -ci permission**

查看ls手册页 **-c** 显示匹配的行数

-i 无视大小写

grep -ni Event main.c **-n** 匹配的行号

Is - List Contents of Directory

* Syntax:

* **ls [-aAbcCdFgfilMnoprRstuxl] [file ...]**

* Frequently used options:

* **-a** all, including files begin with **隐藏文件**

* **-d** directory

* **-l** long

* **-F** format **格式**

* **-R** recursive

* **-t** time

递归所有子目录
时间新到旧

* **compress** [-cfv] [-b bits] [file]

* **uncompress** [-cfv] [file ...] **压缩解小**

* **.Z** as suffix

* **gzip** [-cdhLlnNrtvV19] [-S suffix] [file ...] **中**

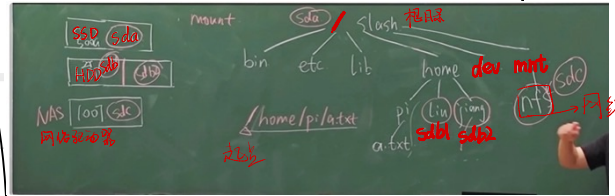
* **gzip filename** filename->filename.gz

* **gzip -d filename** filename.gz->filename

* **gzip -h** help for gzip

* **gunzip**

* **bzip2 / bunzip2** **压缩程度大**



binary 可执行程序目录

etc editable text configuration 系统文件

library 函数库

home 家目录

dev 设备文件(磁盘等)

mnt 临时挂载
往磁盘或网络存储

* **wc** - display a count of lines, words and characters in a file

* **-c**

* **-l**

* **-w**

* **ls | wc -l** 当前目录中 子文件 和目录数量

* **pwd**

```
(jiangjunmin@eex001 ~)$ pwd
/home/jiangjunmin
```

Types Displayed on Linux

* **alias ls** **给ls新定义**

ls --color=auto

blue directories

white/black text files

cyan links

green executables

red compressed archives

pink images

yellow devices

(flashing) red broken links

* **ls -F**

nothing regular file

* executable file

= socket

/ directory

@ link

| named pipe

In and unlink

ln file1 file2

* **ln [options] source target** - Link to another file

* symbolic link **ln -s file1 link**

* **hard link**

* **unlink filename** - Remove the link

find directory-list [options] [actions] [...]

* **find . -name Net*.c -print** **当前目录下**

* **find . -newer empty -print** **比 empty 新**

* **find /usr/local -type d -print** **找 /usr/local 下的所有目录**

* **find . \(user Bill -o -size +10000c\) -print** **Bill 用户 或 大于 10000c (10KB)**

* **find ~Bill ~Denis -size +1000 -atime 30 -ok rm {} \;** **1000 blocks 512KB 30天前 询问 删除**

两个用户的 home

Set User Environment (2)

* **setenv (csh)**

在 csh 中创建或修改环境变量并立即导出给 shell 自动的子进程 **setenv** 变量名 变量值

setenv PATH ~/bin:/usr/bin 不使用符号

setenv MANPATH /usr/share/man:/usr/local/man

setenv MANPATH \$HOME/man:\$MANPATH

setenv **set and export (sh, bash)**

PATH=/usr/local/bin:/home/jack/bin; **export PATH**

export PATH=\$PATH:/opt/acroread/bin

SPATH 表示保留原有路径 **/opt/acroread/bin** 表示在后面追加 Acrobat Reader 的可执行路径

* **alias**

给复杂或常用的命令创建一个短名字 完整命令要用 **'** 包裹起来, 别名不能与系统命令同名 (否则覆盖)

* syntax: **alias** **aliasname** **'command'**

* **alias dir** 'ls -al | more'

dir 这个命令, 执行行为为: 列出所有文件(包括隐藏文件), 用 more 分页显示

* **alias** **list** 'ls -al *.c | more'

```
Linux Bash Shell jiangjunmin@eex001:~
# /bin/bash

# Source global definitions
if [ -f /etc/bashrc ]; then
. /etc/bashrc
fi

# User specific aliases and functions
alias fz='filezilla &'
export PATH=/app/local64/filezilla-3.5.3/bin:/app/local64/rar:/SPATH
alias df='df -h'
alias is='icfb &'
alias epdf='evince'
alias vcs='vcs -full64'

export APP_DIR='/app'

# License file
source /app/config/LICS/bashrc lic
#export LM_LICENSE_FILE=27100@127.0.0.1

# /bin/bash

# Common Path
export PATH=$APP_DIR/matlab/R2011b/bin:$PATH
```

Manual (1)

* man

* Manual sections

* man man

NAME

man - find and display reference manual pages **SYNOPSIS**

man [-] [-adFlrt] [-M path] [-T macro-package] [-s

section] name ...

man [-M path] -k keyword ...

man [-M path] -f file ...

DESCRIPTION

The man command displays information from the reference manuals. It displays complete manual pages that you select

-M 临时覆盖默认路径

Browse through the man Pages

* Spacebar - display next page

* b - previous page

* q - quit

* xman - man has X GUI on Solaris

* KDE/Gnome help browsers on Linux

* Web resources

图形界面

帮助浏览器

符号 / 表达式	核心含义	典型使用场景 (配合命令)
.	当前工作目录 (终端 pwd 命令显示的目录)	- ls .: 列出当前目录下所有内容 - cp /home/file.txt .: 将文件复制到当前目录 - cat ./config.conf: 查看当前目录下的配置文件 (./可简化为.)
..	当前目录的上级目录 (父目录)	- cd ..: 切换到上一级目录 - ls ../docs: 查看上级目录中 docs 文件夹的内容 - mv test.txt ../backup/: 将当前目录文件移到上级目录的 backup 文件夹
./	当前目录的路径前缀 (明确指定“当前目录下”)	- ./start.sh: 执行当前目录下的可执行脚本 (避免与系统命令冲突) - gcc -o app ./main.c: 编译当前目录下的 main.c 文件 - cd ./subdir: 切换到当前目录的 subdir 子目录 (等同于 cd subdir, 更清晰)
~	当前登录用户的家目录 (用户专属默认目录)	- cd ~: 快速回到家目录 (如 /home/ubuntu 或 /root) - ls ~/Downloads: 查看家目录下的 Downloads 文件夹 - touch ~/notes.txt: 在家目录创建 notes.txt 文件
/	系统根目录 (所有目录的最顶层目录)	- cd /: 切换到根目录 - ls /etc: 查看根目录下的 etc 系统配置目录 - cat /var/log/syslog: 查看根目录下的系统日志文件
-	上一次工作目录 (切换目录前的原目录)	- cd -: 在“当前目录”和“上一次目录”之间快速切换 - cp file.txt -: 将文件复制到上一次所在的目录

I²C: 0 通信开始: SCL 保持高电平 并行数据线
SDA 从高位到低位 ~ 时钟 ~
↓
start condition 表示通信开始

M 发出 7 位地址 + { 0 主从从入
地址后是应答位(Ack), 从设备
根据其地址进行识别并向主~
发送应答信号

Δ I²C 每次传 8 位 (1 字节), 接收后会发一个
应答位

写: 主→从
读: 从→主

line Manual

)

OPTIONS The following options are supported:

-a Show all manual pages matching name within the MANPATH search path. Manual pages are displayed in the order found.

-d Debug. Displays what a section-specifier evaluates to, method used for searching, and paths searched by man.

-f file...

man attempts to locate manual pages related to any of the given file s. It strips the leading path name components from each file, and then prints one-line summaries containing the resulting basename or names. This option also uses the win-dex database.....

er Paragraphs in 'man

in'

man man

man 命令手册的末尾部分

USAGE

ENVIRONMENT VARIABLES

EXIT STATUS

SEE ALSO

apropos(1), cat(1), col(1), eqn(1), more(1), nroff(1), refer(1), tbl(1), troff(1), vgrind(1), whatis(1), catman(1M), attributes(5), environ(5), eqnchar(5), man(5), sgml(5)

BUGS

Directory Commands

(2) (在上版)

* mkdir directory-list

* Example: mkdir personal manage

创建了 2 个目录

* rmdir directory-list

* Directory must be empty

* Example: rmdir personal manage

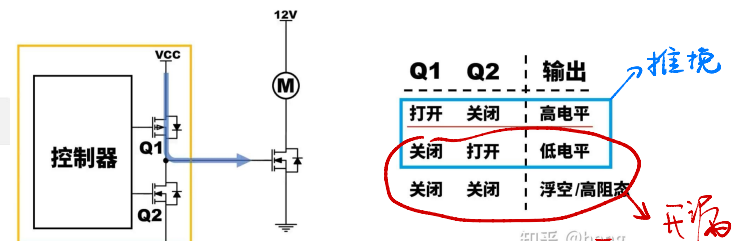
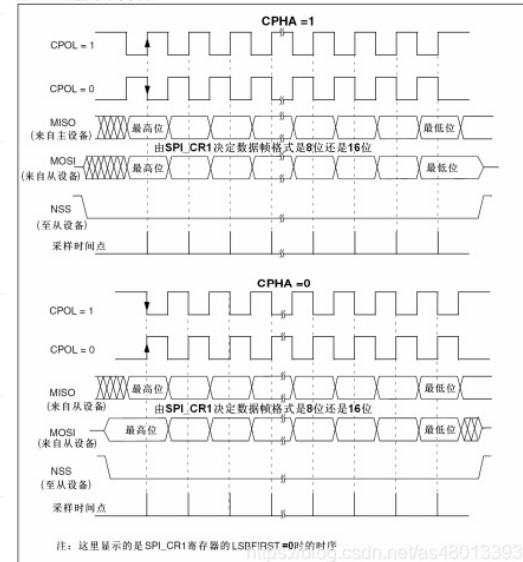


图212 数据时钟时序图



7位 I²C 9x00-0x0F 禁用 2⁷-1=127
0x78-0x7F

修改文件或目录的访问权限(read/write/execute)

* chmod change the file or directory access permissions (mode)

* umask get or set the file mode creation mask

查看或设置新建文件或目录默认权限的掩码

mask 新文件权限=默认权限-umask

改变文件或目录的所属用户组 需要有写权限或 root 权限

* chgrp change the group of the file

改变文件或目录的拥有者 普通修改自己,root 修改任意

* chown change the owner of a file

List Directory Contents

File Structures on Disk

磁盘被挂载到文件树的某个位置-Linux 只有一个统一的文件树

不同的磁盘、分区不会像 Windows 那样出现 C:、D:

* Disks mounted to positions in file tree

磁盘最小的读写单位不是文件,而是块,常见块大小:512bytes(1KB)

* A disk is divided into blocks

* 512, 1024, 2048 bytes are common block sizes

* A disk partition divided into three major sections
超级重要的小块区域,包含:文件系统类型(ext4,xfs),block 大小,总块数,inode 数量;被使用/空闲的superblock 块数量;文件系统的状态(是否正常关闭) 如果环境,文件系统无法识别每个文件都对应一个 inode(索引节点),inode 里面存放:文件的元数据:拥有者;权限;创建时间;修改时间;inode 间:文件大小;指向数据块的地址(文件数据在哪里) 不存在文件名
真正存储文件内容的地方,比如:文本内容,图片数据,可执行程序的二进制,文件最终以块的形式存储在 data blocks

* More left for later on System Administration

[illegible]

图 1-1 展示了树莓派4B的板载接口。图中是一个绿色的电路板，上面标有“Raspberry Pi 4 Model B”字样。接口包括：顶部的双频WiFi/蓝牙5.0天线、内存芯片、网卡、PoE接口、千兆以太网端口、USB 3.0和USB 2.0端口；左侧的DSI显示连接器；底部的Type-C供电接口、micro HDMI接口、摄像头接口、音频接口和USB管理芯片。

```
piano% who
opc          console Apr 16 08:02 (:0)
liangzs     pts/2    Apr 16 15:49 (10.14.111.198)
yu jun      dtremote Apr 18 19:37 (10.14.111.244:0)
wangs j     pts/10    Apr 18 12:47 (10.14.111.132)
```

```

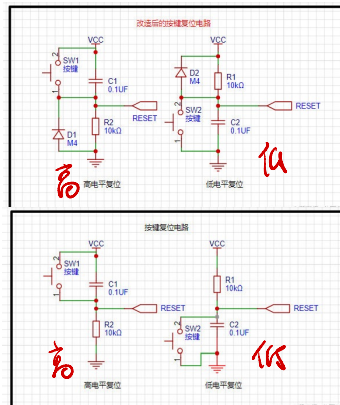
pi@no% ps
PID TTY TIME CMD
4059 pts/38 0:00 csh
4061 pts/38 0:01 ps
4286 pts/38 0:00 xterm

```

判断文件类型

- * **file** - determine file type
- * **sort** - sort, merge, or sequence check text files
- * **which** - locate a command; display its pathname or alias
- * **alias** - set command aliases
- * **whatis** - search the whatis database for word
- * **word** - search the whatis database for string
- * **spell** - report spelling errors of file
- * **look** - find words in dictionary
- * Other standard utilities in /usr/bin, /usr/sbin

- * **stty** 作用:查询或设置当前终端的行程码参数,比如按哪个键删除
→ **stty - terminal control** 修改,按哪个键调整行
- 终端是在内核里的 **tty** 行程码, **stty** 是用户空间符合这条规则的唯一命令
- * reports or sets terminal control options
configures aspects of I/O control
- * syntax: **stty attribute value**
- * example: stty erase ^H; stty kill ^U;
变量分shell 变量与环境变量;只有 export 后的才会被 fork 出
- * **Environment variables**, 的子进程继承
login 程序从/etc/passwd 第6字段读取用户的环境。
- * **HOME** 影响: cd 不带参数就进 HOME; 很多程序 (vim、bash、git) 默认在这里找配置 dot-file, 手动改找不到则配置
- * **PATH** env 环境变量,搜索到可执行文件。
推荐语法是: PATH=HOME/bin:\$PATH 保证个人脚本优先
- * **SHELL** ,又保留系统路径。
login 程序根据/etc/passwd 最后一字段填写。
- * **env - show current Env values**
排查: 登录后 env | grep -P '^PATH='HOME 查看是否被安全外置。
清理: env | bash -l 启动一个环境的全新 bash,方便测试。



+ 关注

- 总线电容 C_b
 - 总线电容 (C_b) = 线缆电容 (C_w) + 连接电容 (C_c) + 引脚电容 (C_i)
 - 上拉电阻的最大值, 和总线电容决定了信号上升和下降的时间。
- $V_{IL} = 0.3V_{DD}$, $V_{IH} = 0.7V_{DD}$
 - $0.3 \times V_{DD} = V_{DD} (1 - e^{-\frac{t_1}{RC_b}})$, $t_1 = 0.3566749 \times RC_b$
 - $0.7 \times V_{DD} = V_{DD} (1 - e^{-\frac{t_2}{RC_b}})$, $t_2 = 1.2039729 \times RC_b$
 - $t_r = t_2 - t_1 = 0.8473 \times RC_b$
- $R_{P(max)} = \frac{t_r}{0.8473 \times C_b}$
 - $t_r = 1000\text{ns}$ (标准模式), 300ns (快速模式), 120ns (快速模式+)
 - C_b 是一个对总线电容的估算值。

最小值

$$\frac{V_{DD} - V_{OLmax}}{I_{OL}} = \frac{3.3V - 0.4V}{3mA}$$